

DIVYA SINGH

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EDUCATION

Rishihood University, Sonapat

Aug 2024 – May 2028

Bachelor of Technology in Computer Science and Data Science

CGPA: 8.72

Relevant Coursework: **Object-Oriented Programming (OOP)**, **Data Structures and Algorithms**, **Operating Systems (OS)**, **Computer Networks (CN)**, **Database Management Systems (DBMS)**

EXPERIENCE

Beyond Innovation

Noida, Uttar Pradesh

Software Engineer Intern

May 2026 – July 2026

- Developed client-facing frontend features and integrated **REST APIs** for complex data workflows, contributing to a **40% improvement** in application data reliability.
- Collaborated with **backend engineers** to improve **production data synchronization**, performing **code reviews** and implementing technical feedback to reduce synchronization errors.
- Identified **production performance bottlenecks** and shipped optimized **pull requests**, reducing average response latency by **25%**.

PROJECTS

ReMi – RAG Research Agent with Hallucination Grounding | *Python, FAISS, spaCy, Gemini, Sentence-Transformers*

[GitHub](#) | [Live](#)

- Engineered a **5-stage autonomous research pipeline** using **Gemini Flash**, generating evidence-driven reports by decomposing user queries into **3–5 sub-questions**.
- Built a local **RAG pipeline** using **FAISS** and MiniLM embeddings with hallucination grounding, computing **Faithfulness** and **Coverage** scores against retrieved evidence.
- Optimized retrieval performance by identifying and resolving an $O(N \log N)$ retrieval bottleneck and an $O(N \cdot D)$ index reconstruction cost, while maintaining a **100% pass rate** across **20 Pytest unit tests**.

CollabDocs – Real-Time Collaborative Document Editor | *React, Node.js, Express.js, Socket.io, PostgreSQL, Prisma*

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- Built a real-time collaborative editing engine by implementing a custom **Operational Transformation** algorithm and a **Socket.io** synchronization layer handling **8 event types**.
- Designed and implemented **12 REST API endpoints** for authentication, document management, and secure sharing using **Google OAuth** and **JWT**.
- Optimized concurrent editing performance by eliminating a full-document broadcast bottleneck and improving PostgreSQL query efficiency through proper foreign-key indexing.

ShopSmart – Cloud Deployment Pipeline & Infrastructure | *Terraform, Docker, AWS ECS Fargate, GitHub Actions, Node.js, React*

[GitHub](#)

- Provisioned cloud infrastructure using **Terraform**, deploying **12 AWS resources** including ECS Fargate, ECR, S3, and CloudWatch logging.
- Built a **3-stage CI/CD pipeline** using GitHub Actions, automating testing, infrastructure provisioning, and zero-downtime deployments to ECS.
- Diagnosed an unmemoized React component tree causing cascading re-renders during cart interactions, identifying **React.memo** and state decoupling as the optimization strategy.

ACHIEVEMENTS

- Selected as Institutional Nominee at Smart India Hackathon 2024** for engineering a real-time, scalable gunshot detection system under competitive constraints.
- Solved **500+ DSA problems** across **LeetCode**, **CodeChef**, **Codeforces**, and **GeeksforGeeks**, strengthening proficiency in algorithms and data structures.
- Attained a peak **LeetCode** rating of **1546**, earned a **2-Star** rating on **CodeChef**, and achieved **Pupil** rank on **Codeforces** with a peak rating of **1222** (Global Rank **1342** in a Div. 2 contest).

SKILLS

- Core CS:** DSA, OOP, DBMS, OS, Computer Networks, Distributed Systems, System Design, Operational Transformation
- Languages:** C++, JavaScript, TypeScript, Python, SQL
- Backend:** Node.js, Express.js, FastAPI, REST APIs, Microservices, Socket.io, Prisma, JWT, OAuth 2.0
- Frontend:** React.js, Next.js, Tailwind CSS
- Databases:** PostgreSQL, MongoDB, MySQL, Redis
- Cloud & DevOps:** Docker, Terraform, GitHub Actions, CI/CD, AWS (ECS Fargate, ECR, S3, CloudWatch)
- AI/ML & Gen AI:** NLP, NumPy, Pandas, spaCy, Sentence Transformers, RAG Pipelines, Gemini API, LangChain, FAISS, Prompt Engineering
- Tools & Testing:** Git, GitHub, Postman, Pytest, Unit Testing